Ensemble-Based Spatial Snow Water Equivalent Estimates Using the SWE Evaluation Tool for the Rio Grande Basin

February 8, 2026

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Introduction

No spaceborne sensor can directly measure snow water equivalent (SWE). Therefore, the spatial distribution of SWE must be estimated from remotely sensed observations combined with modeling approaches. Existing gridded SWE products exhibit documented accuracy limitations across both mountainous and non-mountainous regions, resulting in a wide range of SWE estimates for any given location. To address this uncertainty, the New Mexico Office of the State Engineer (NM OSE), in collaboration with the Institute of Arctic and Alpine Research (INSTAAR), developed the SWE Evaluation Tool (SWEET). SWEET integrates multiple suitable SWE data products into an ensemble framework, with the objective of producing near-real-time spatial SWE estimates for the Rio Grande Basin that are, on average, as accurate as or more accurate than any individual product alone.

Figure 1 shows the near-real-time (February 8, 2026) ensemble SWE estimates for the Rio Grande Basin. Subsequent sections of this report present detailed ensemble-based SWE maps, along with summaries by subbasin and elevation band. In addition to the ensemble results highlighted in this report, figures and tables for each individual SWE product are also provided for download to support transparency and further analysis. These data sets and visualizations are available on the GitHub repository.

For ease of navigation, hyperlinks are provided throughout this report to help readers quickly access basin-specific sections and supporting data tables.

Rio Grande Basin SWE Overview

Rio Grande Headwaters

Upper Rio Grande

Rio Grande-Elephant Butte (Middle Rio Grande)

About SWEET

Rio Grande Basin SWE Overview

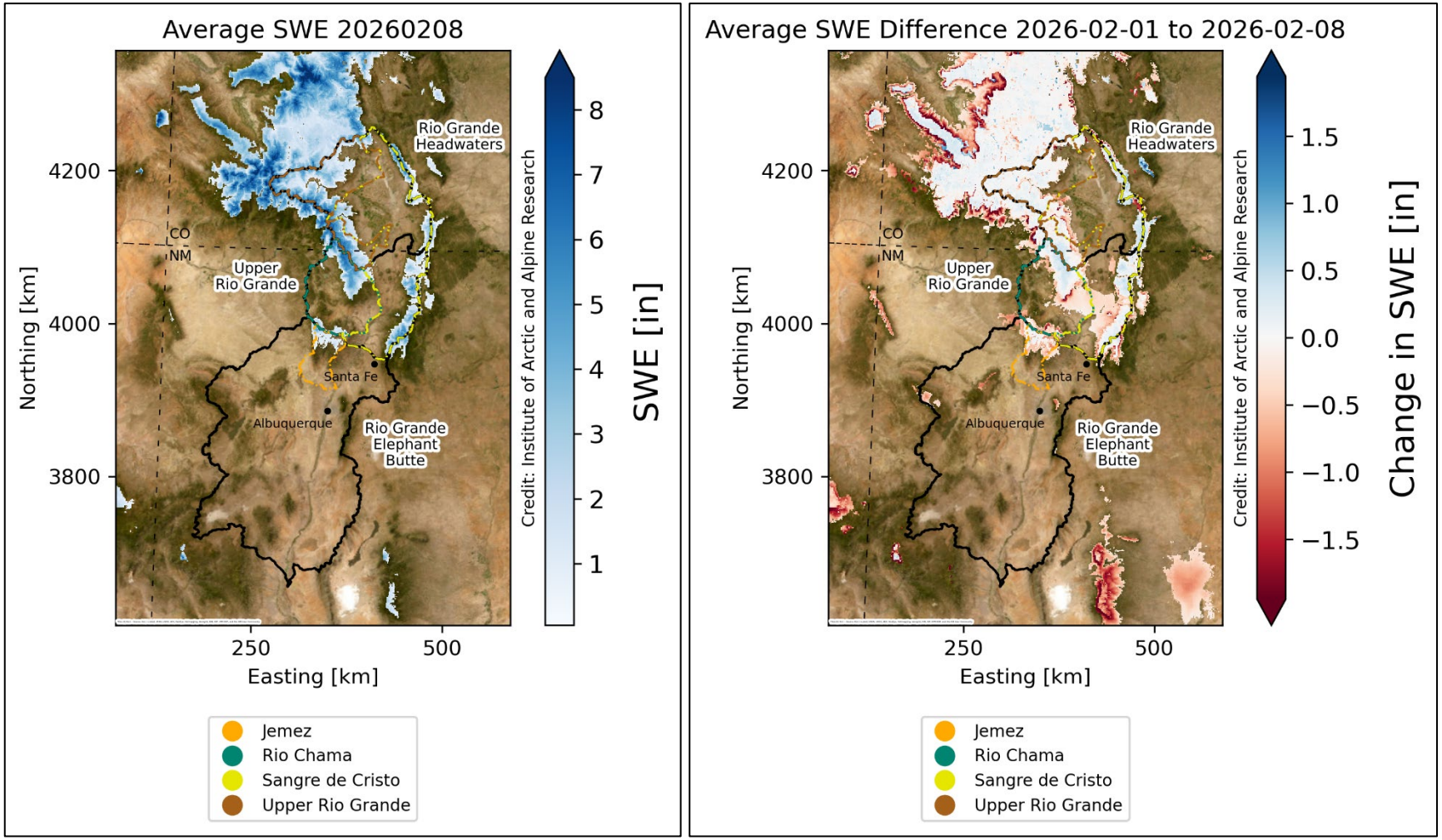

Figure 1. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Basin for February 8, 2026. *SWE amounts (in.) across the Rio Grande Basin (left); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Basin from February 1, 2026, to February 8, 2026 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Table 1. Estimated ensemble SWE by HUC6 Basin for February 8, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 1, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-01 (af)	Median Vol. Change since 2026-02-01 (af)
Rio Grande Headwaters	Rio Grande Headwaters	203,414	205,937	192,227 (UA_SWE)	212,078 (CU_SWE)	1,433	5,081
	Alamosa-Trinchera	78,415	77,742	68,134 (SNODAS)	89,369 (UA_SWE)	-7,721	-9,648
	San Luis	50,158	49,298	44,026 (SNODAS)	57,151 (UA_SWE)	-3,613	-3,217
	Saguache	56,311	49,634	46,337 (SNODAS)	72,961 (UA_SWE)	-5,425	-12,781
	Conejos	100,060	99,225	86,265 (CU_SWE)	114,690 (SNODAS)	-187	-4,333
	Total	**488,358**	**479,124**	**475,017 (CU_SWE)**	**510,933 (UA_SWE)**	**-15,513**	**-19,133**
Upper Rio Grande	Upper Rio Grande	95,836	116,809	47,166 (CU_SWE)	123,534 (SNODAS)	-20,090	2,228
	Rio Chama	134,320	125,707	109,486 (CU_SWE)	167,766 (SNODAS)	-20,544	-34,833
	Total	**230,156**	**242,516**	**156,653 (CU_SWE)**	**291,300 (SNODAS)**	**-40,633**	**-32,605**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	1,574	1,582	1,085 (CU_SWE)	2,056 (SNODAS)	-3,970	-3,252
	Jemez	16,571	12,864	11,610 (CU_SWE)	25,239 (UA_SWE)	-6,600	-10,765
	Rio Grande-Albuquerque	184	138	12 (SNODAS)	400 (CU_SWE)	-1,262	-790
	Arroyo Chico	1,367	25	0 (SNODAS)	4,074 (CU_SWE)	-3,554	-2,513
	North Plains	0	0	0	0	-985	-941
	Rio San Jose	878	286	6 (SNODAS)	2,342 (CU_SWE)	-6,823	-6,800
	Plains of San Agustin	61	28	3 (UA_SWE)	154 (CU_SWE)	-582	-382
	Rio Salado	0	0	0	0	-10	-7
	Jornada Del Muerto	0	0	0	0	-38	-12
	Elephant Butte Reservoir	188	2	0 (UA_SWE)	563 (CU_SWE)	-718	-368
	Rio Puerco	4,362	2,996	1,973 (CU_SWE)	8,117 (UA_SWE)	-5,139	-6,826
	Total	**25,186**	**22,201**	**17,964 (SNODAS)**	**35,392 (UA_SWE)**	**-29,681**	**-47,938**

Table 2. Estimated ensemble SWE by specified subbasin groupings for February 8, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 1, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

OSE Grouping	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-01 (af)	Median Vol. Change since 2026-02-01 (af)
Upper Rio Grande	Rio Grande Headwaters	203,414	205,937	192,227 (UA_SWE)	212,078 (CU_SWE)	1,433	5,081
	Saguache	56,311	49,634	46,337 (SNODAS)	72,961 (UA_SWE)	-5,425	-12,781
	Conejos	100,060	99,225	86,265 (CU_SWE)	114,690 (SNODAS)	-187	-4,333
	Total	**359,785**	**364,413**	**347,977 (CU_SWE)**	**366,964 (SNODAS)**	**-4,180**	**5,756**
Sangre de Cristo	San Luis	50,158	49,298	44,026 (SNODAS)	57,151 (UA_SWE)	-3,613	-3,217
	Alamosa-Trinchera	78,415	77,742	68,134 (SNODAS)	89,369 (UA_SWE)	-7,721	-9,648
	Upper Rio Grande	95,836	116,809	47,166 (CU_SWE)	123,534 (SNODAS)	-20,090	2,228
	Total	**224,410**	**235,694**	**174,206 (CU_SWE)**	**263,328 (UA_SWE)**	**-31,423**	**21,990**
Rio Chama	**Total**	**134,320**	**125,707**	**109,486 (CU_SWE)**	**167,766 (SNODAS)**	**-20,544**	**-34,833**
Jemez	**Total**	**16,571**	**12,864**	**11,610 (CU_SWE)**	**25,239 (UA_SWE)**	**-6,600**	**-10,765**

Upper Rio Grande Snowpack Update Application

The OSE/ISC Upper Rio Grande Snowpack Update Application (Application) provides water managers and users with detailed SNOTEL-based SWE data organized by the same subbasin groupings shown in Table 2. The Application is updated regularly from December through May and compiles key SWE information from SNOTEL sites within each subbasin. In addition to the Upper Rio Grande groupings (Upper Rio Grande, Sangre de Cristo, Rio Chama, and Jemez), the Application also includes the San Juan River Basin, with data summarized for the Upper San Juan, Animas, and Piedra subbasins.

The Application can be viewed here[1].

[1] Please note that access to the Application may be restricted to employees of the NM OSE/ISC at this time. We are currently working on public access.

Rio Grande Headwaters

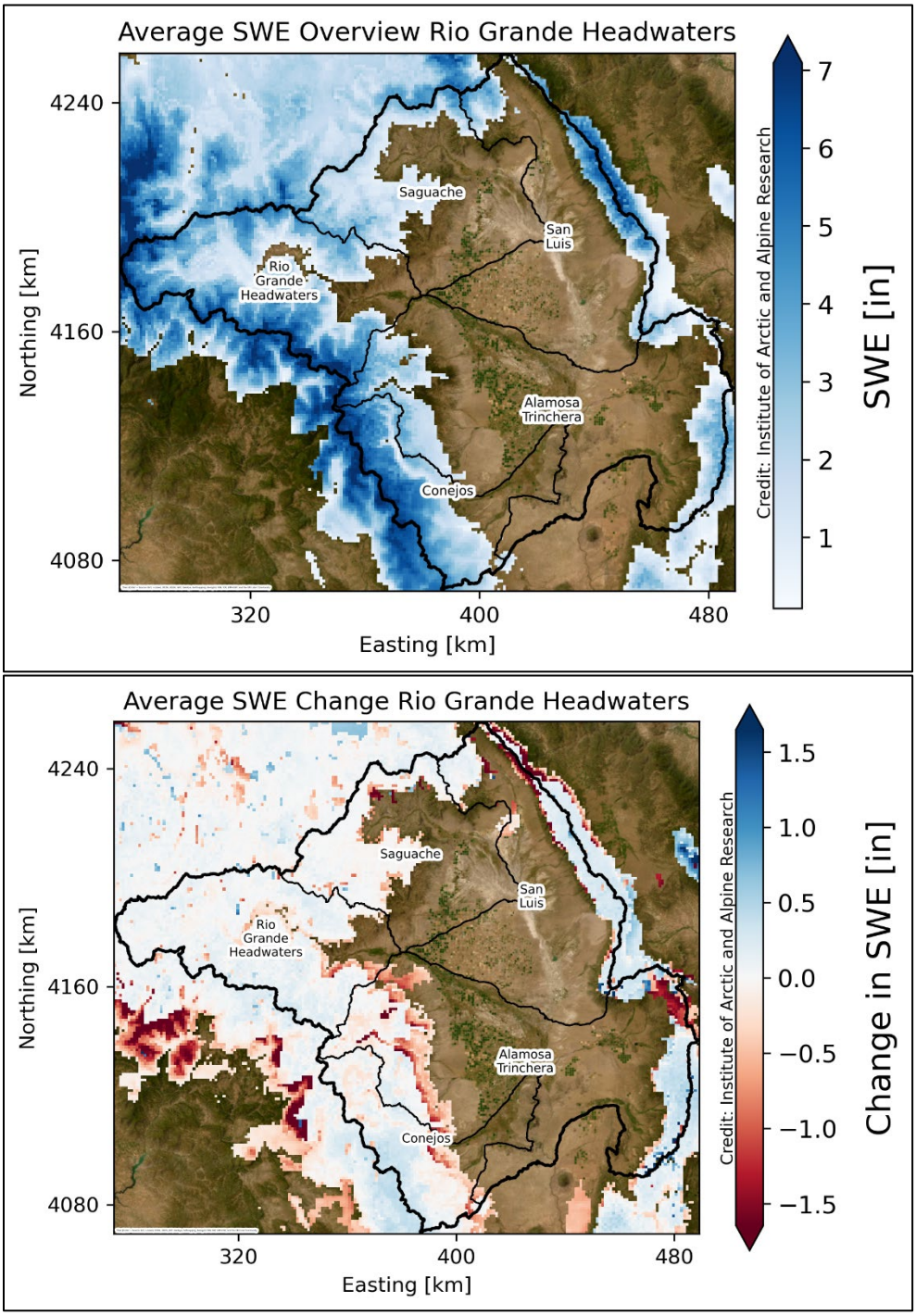

Figure 2. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Headwaters HUC6 Basin for February 8, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande Headwaters Basin, showing spatial patterns of SWE across subbasins including Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Headwaters from February 1, 2026, to February 8, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

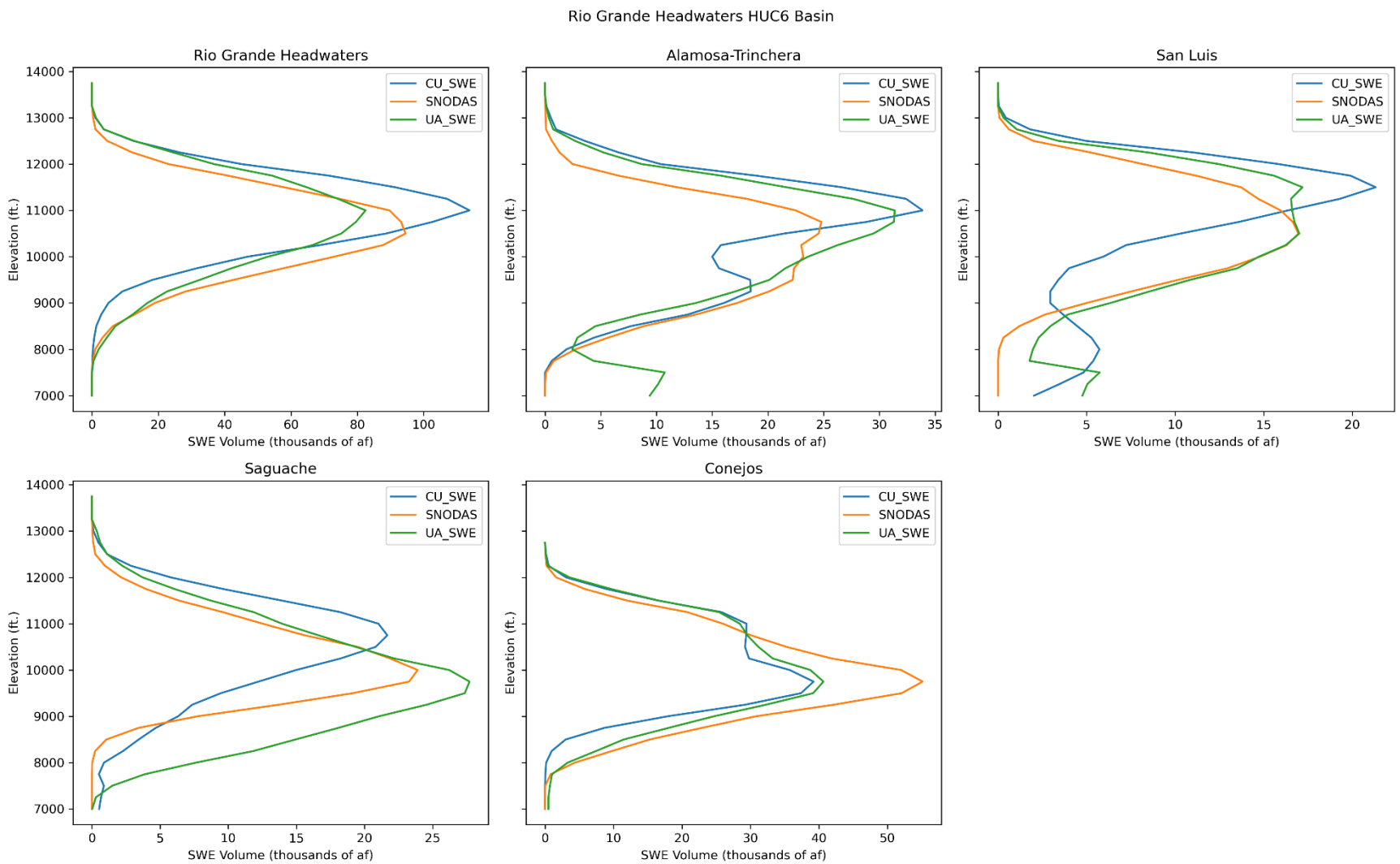

Figure 3. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for February 8, 2026. *Elevation-distributed SWE volume for the Rio Grande Headwaters HUC6 Basin and associated subbasins (Rio Grande Headwaters, Alamosa–Trinchera, San Luis, Saguache, and Conejos), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 3. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for February 8, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 1, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-01 (af)	Median Vol. Change since 2026-02-01 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0	0	0	0	-21	-18
		8000-9000'	1,087	964	288 (CU_SWE)	2,010 (UA_SWE)	-1,120	-1,333
		9000-10000'	13,525	16,698	4,953 (CU_SWE)	18,925 (SNODAS)	-620	-545
		10000-11000'	57,643	53,046	46,895 (CU_SWE)	72,988 (SNODAS)	749	-262
		11000-12000'	95,326	89,669	82,482 (UA_SWE)	113,826 (CU_SWE)	2,001	2,189
		12000-13000'	34,980	36,806	23,048 (SNODAS)	45,084 (CU_SWE)	432	-462
		13000-14000'	853	1,032	341 (SNODAS)	1,186 (UA_SWE)	11	41
	Alamosa-Trinchera	7000-8000'	3,133	0	0 (SNODAS, CU_SWE)	9,400 (UA_SWE)	-2,953	-923
		8000-9000'	2,363	2,438	1,923 (CU_SWE)	2,729 (SNODAS)	-5,369	-6,420
		9000-10000'	15,608	16,052	13,548 (UA_SWE)	17,223 (SNODAS)	-1,592	-2,014
		10000-11000'	20,591	23,175	15,002 (CU_SWE)	23,597 (UA_SWE)	1,024	3,150
		11000-12000'	29,244	31,380	22,476 (SNODAS)	33,875 (CU_SWE)	819	310
		12000-13000'	7,169	8,667	2,475 (SNODAS)	10,364 (CU_SWE)	327	93
		13000-14000'	307	339	57 (SNODAS)	524 (CU_SWE)	23	1
	San Luis	7000-8000'	2,263	2,030	0 (SNODAS)	4,760 (UA_SWE)	-3,618	-1,246
		8000-9000'	2,580	1,966	50 (SNODAS)	5,723 (CU_SWE)	-1,341	-2,591
		9000-10000'	4,759	4,996	2,946 (CU_SWE)	6,336 (UA_SWE)	-420	640
		10000-11000'	11,815	14,724	5,938 (CU_SWE)	14,782 (SNODAS)	362	1,620
		11000-12000'	16,309	16,376	15,949 (SNODAS)	16,603 (UA_SWE)	716	434
		12000-13000'	12,159	12,459	8,173 (SNODAS)	15,844 (CU_SWE)	670	24
		13000-14000'	273	303	76 (SNODAS)	441 (CU_SWE)	18	-1
	Saguache	7000-8000'	182	19	0 (SNODAS)	527 (CU_SWE)	-894	-1,053
		8000-9000'	2,842	878	14 (SNODAS)	7,635 (UA_SWE)	-904	-1,877
		9000-10000'	11,697	7,723	6,318 (CU_SWE)	21,052 (UA_SWE)	-2,887	-6,247
		10000-11000'	21,702	23,900	14,983 (CU_SWE)	26,223 (UA_SWE)	-1,036	632
		11000-12000'	15,839	13,954	12,538 (SNODAS)	21,025 (CU_SWE)	213	-45
		12000-13000'	3,884	3,723	2,139 (SNODAS)	5,790 (CU_SWE)	82	-31
		13000-14000'	164	113	24 (SNODAS)	354 (UA_SWE)	1	7
	Conejos	7000-8000'	161	0	0 (SNODAS, CU_SWE)	484 (UA_SWE)	-334	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-01 (af)	Median Vol. Change since 2026-02-01 (af)
Rio Grande Headwaters	Conejos	8000-9000'	2,606	3,301	167 (CU_SWE)	4,349 (SNODAS)	-1,047	-1,410
		9000-10000'	24,427	24,650	17,874 (CU_SWE)	30,758 (SNODAS)	-757	-1,201
		10000-11000'	42,160	38,749	35,733 (CU_SWE)	51,997 (SNODAS)	1,739	-46
		11000-12000'	27,950	28,451	25,974 (SNODAS)	29,424 (CU_SWE)	217	753
		12000-13000'	2,757	3,067	1,612 (SNODAS)	3,590 (UA_SWE)	-5	134

Upper Rio Grande

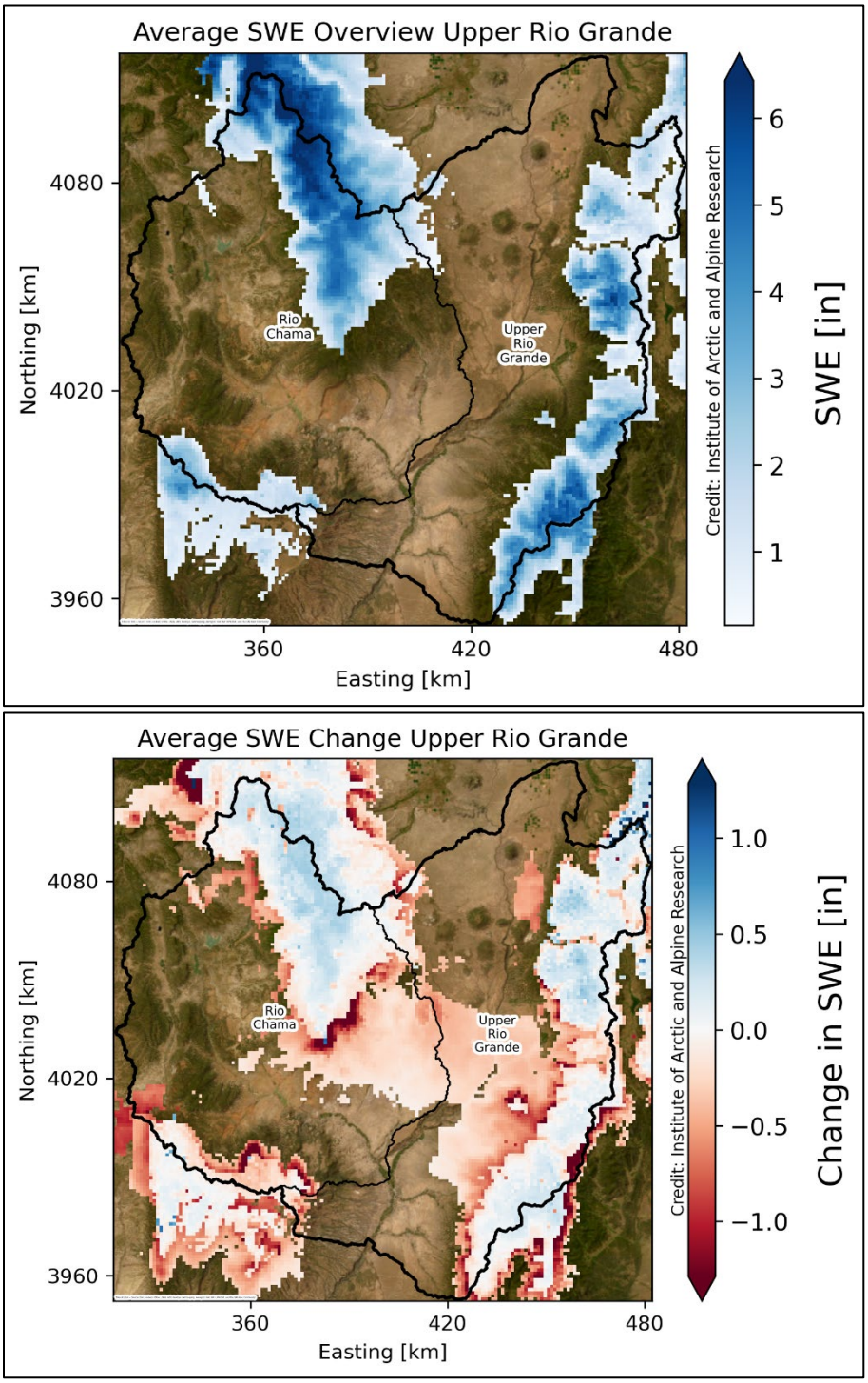

Figure 4. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Upper Rio Grande HUC6 Basin for February 8, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Upper Rio Grande Basin, showing spatial patterns of SWE across subbasins including Rio Chama and Upper Rio Grande (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Upper Rio Grande Basin from February 1, 2026, to February 8, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

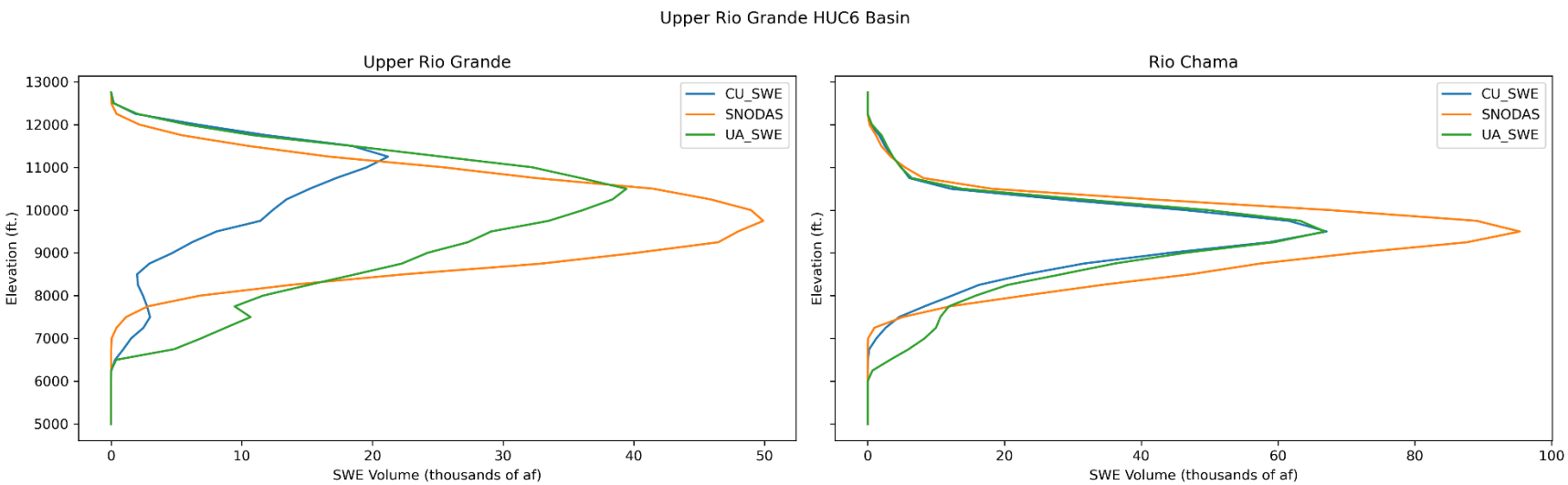

Figure 5. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Upper Rio Grande HUC6 Basin for February 8, 2026. *Elevation-distributed SWE volume for the Upper Rio Grande HUC6 Basin and associated subbasins (Rio Chama and Upper Rio Grande), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 4. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Upper Rio Grande HUC6 Basin for February 8, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 1, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-01 (af)	Median Vol. Change since 2026-02-01 (af)
Upper Rio Grande	Upper Rio Grande	5000-6000'	0	0	0	0	-27	0
		6000-7000'	0	0	0	0	-1,943	-1,277
		7000-8000'	2,805	1,522	33 (SNODAS)	6,859 (UA_SWE)	-10,939	-5,676
		8000-9000'	6,941	6,773	2,439 (CU_SWE)	11,613 (UA_SWE)	-5,697	-1,154
		9000-10000'	23,015	24,196	4,697 (CU_SWE)	40,153 (SNODAS)	-2,818	-7,450
		10000-11000'	32,478	36,102	12,361 (CU_SWE)	48,970 (SNODAS)	109	-2,008
		11000-12000'	25,751	25,451	19,560 (CU_SWE)	32,240 (UA_SWE)	955	3,632
		12000-13000'	4,847	5,799	2,155 (SNODAS)	6,587 (CU_SWE)	270	-13
	Rio Chama	5000-6000'	0	0	0	0	-17	0
		6000-7000'	0	0	0 (SNODAS, CU_SWE)	1 (UA_SWE)	-2,271	-1,041
		7000-8000'	3,209	1,262	59 (SNODAS)	8,306 (UA_SWE)	-15,235	-12,189
		8000-9000'	16,990	15,797	12,318 (CU_SWE)	22,855 (SNODAS)	-7,055	-5,787
		9000-10000'	53,948	46,315	44,086 (CU_SWE)	71,443 (SNODAS)	790	-5,465
		10000-11000'	54,665	49,814	46,489 (CU_SWE)	67,692 (SNODAS)	3,112	146
		11000-12000'	5,075	4,927	4,843 (UA_SWE)	5,454 (SNODAS)	121	106
		12000-13000'	433	406	262 (SNODAS)	631 (UA_SWE)	10	22

Rio Grande-Elephant Butte (Middle Rio Grande)

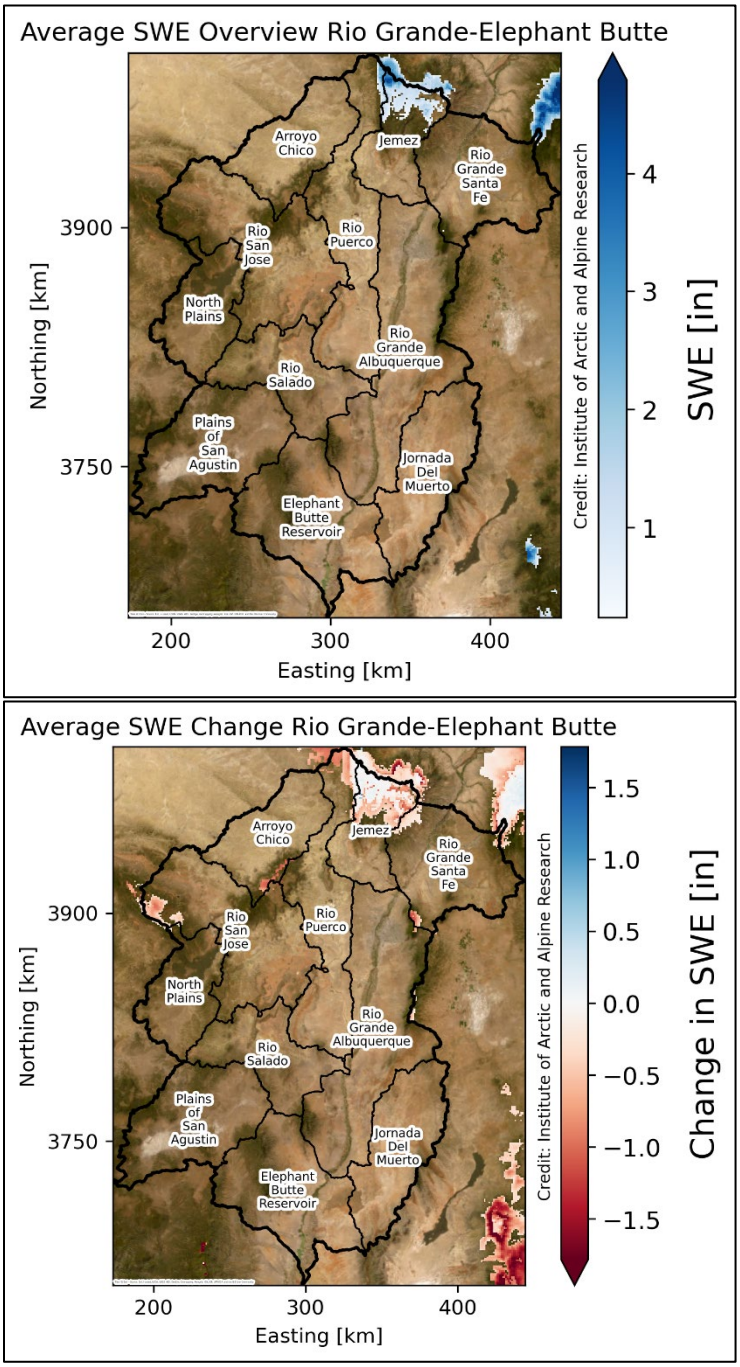

Figure 6. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande-Elephant Butte HUC6 Basin for February 8, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande-Elephant Butte Basin, showing spatial patterns of SWE across subbasins including Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande-Elephant Butte Basin from February 1, 2026, to February 8, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

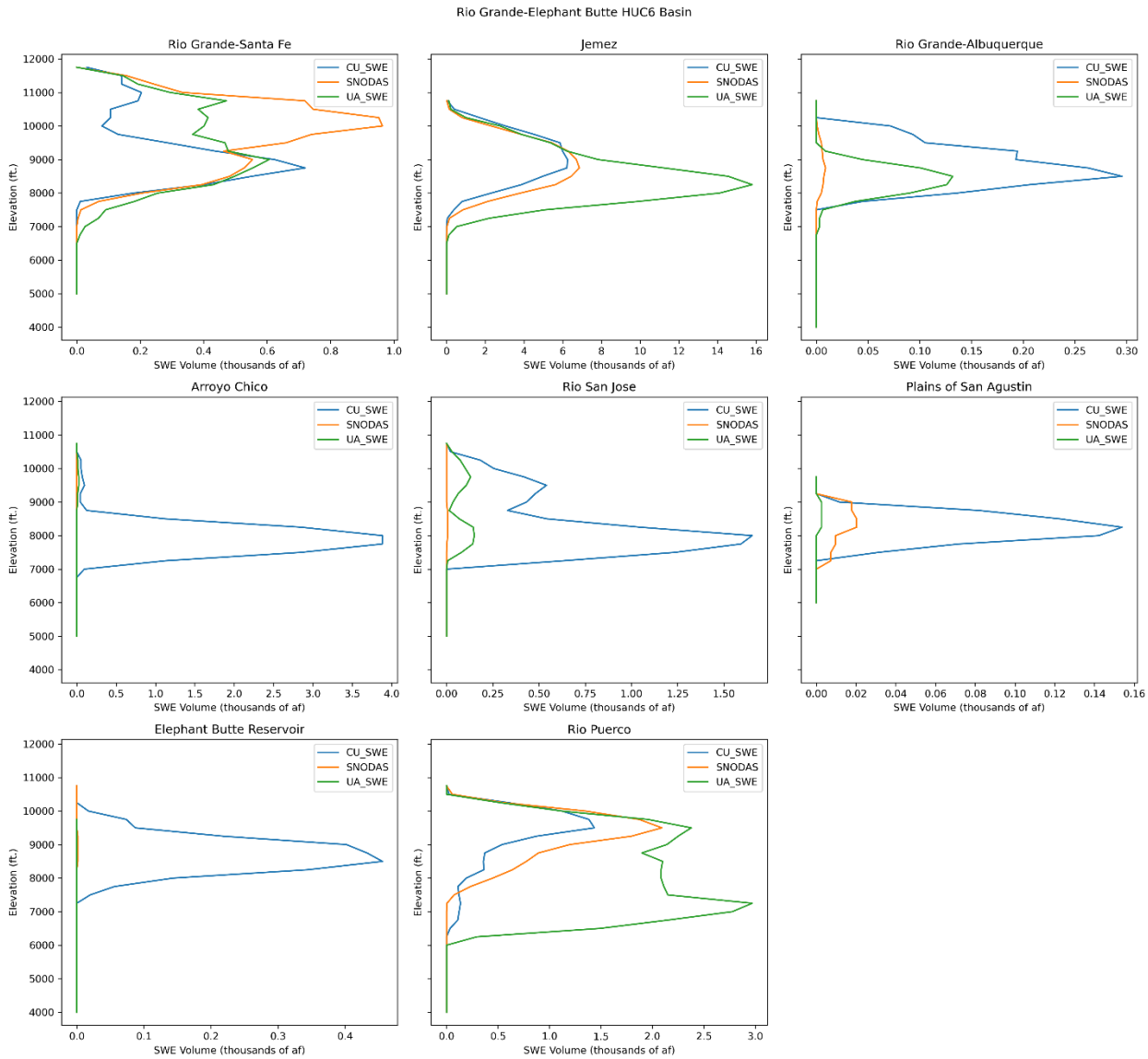

Figure 7. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for February 8, 2026. *Elevation-distributed SWE volume for the Rio Grande-Elephant Butte HUC6 Basin and associated subbasins (Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado[2], North Plains[2], Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto[2]), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

[2] HUC8 subbasins not included because no volume was detected by any SWE products.

Table 5. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for February 8, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 1, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-01 (af)	Median Vol. Change since 2026-02-01 (af)
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0	0	0	0	-28	-1
		6000-7000'	0	0	0	0	-947	-1,106
		7000-8000'	9	0	0 (SNODAS, CU_SWE)	26 (UA_SWE)	-1,166	-528
		8000-9000'	213	206	179 (CU_SWE)	255 (UA_SWE)	-1,328	-1,453
		9000-10000'	594	606	554 (SNODAS)	623 (CU_SWE)	-480	-624
		10000-11000'	481	401	79 (CU_SWE)	963 (SNODAS)	-18	-74
		11000-12000'	277	294	204 (CU_SWE)	333 (SNODAS)	-4	-14
	Jemez	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-36	0
		7000-8000'	181	21	0 (CU_SWE)	522 (UA_SWE)	-932	-610
		8000-9000'	6,749	3,822	2,310 (CU_SWE)	14,113 (UA_SWE)	-3,870	-5,934
		9000-10000'	6,932	6,709	6,255 (CU_SWE)	7,833 (UA_SWE)	-1,660	-2,801
		10000-11000'	2,709	2,771	2,312 (SNODAS)	3,044 (CU_SWE)	-102	-239
	Rio Grande-Albuquerque	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-80	-92
		7000-8000'	1	0	0 (SNODAS, CU_SWE)	3 (UA_SWE)	-480	-159
		8000-9000'	77	90	5 (SNODAS)	136 (CU_SWE)	-483	-253
		9000-10000'	81	45	6 (SNODAS)	193 (CU_SWE)	-199	-247
		10000-11000'	24	1	0 (UA_SWE)	72 (CU_SWE)	-20	-20
	Arroyo Chico	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-154	-168
		7000-8000'	32	0	0 (SNODAS, UA_SWE)	95 (CU_SWE)	-1,724	-1,234
		8000-9000'	1,294	0	0 (SNODAS, UA_SWE)	3,881 (CU_SWE)	-1,613	-830
		9000-10000'	19	11	0 (SNODAS)	44 (CU_SWE)	-54	-51
		10000-11000'	22	14	0 (SNODAS)	53 (CU_SWE)	-9	-11
	North Plains	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-640	-694
		8000-9000'	0	0	0	0	-346	-246

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-01 (af)	Median Vol. Change since 2026-02-01 (af)
Rio Grande-Elephant Butte	Rio San Jose	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-4	0
		7000-8000'	0	0	0	0	-792	-1,141
		8000-9000'	603	150	6 (SNODAS)	1,653 (CU_SWE)	-5,578	-4,951
		9000-10000'	156	35	0 (SNODAS)	433 (CU_SWE)	-230	-525
		10000-11000'	119	102	0 (SNODAS)	256 (CU_SWE)	-219	-145
	Plains of San Agustin	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-7	-2
		8000-9000'	51	10	0 (UA_SWE)	142 (CU_SWE)	-492	-323
		9000-10000'	11	12	3 (UA_SWE)	18 (SNODAS)	-83	-92
	Rio Salado	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-2	0
		7000-8000'	0	0	0	0	-1	0
		8000-9000'	0	0	0	0	-4	0
		9000-10000'	0	0	0	0	-3	0
	Jornada Del Muerto	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	-1	0
		6000-7000'	0	0	0	0	-19	-7
		7000-8000'	0	0	0	0	-17	-10
		8000-9000'	0	0	0	0	-2	-2
	Elephant Butte Reservoir	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-3	-2
		7000-8000'	0	0	0	0	-9	-13
		8000-9000'	48	0	0 (SNODAS, UA_SWE)	143 (CU_SWE)	-137	-102
		9000-10000'	135	2	0 (UA_SWE)	402 (CU_SWE)	-569	-287
		10000-11000'	9	9	0 (SNODAS)	17 (CU_SWE)	-0	-0
	Rio Puerco	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-347	-136
		7000-8000'	966	122	0 (SNODAS)	2,776 (UA_SWE)	-3,547	-5,671
		8000-9000'	908	448	191 (CU_SWE)	2,084 (UA_SWE)	-1,210	-1,776

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-01 (af)	Median Vol. Change since 2026-02-01 (af)
Rio Grande-Elephant Butte	Rio Puerco	9000-10000'	1,292	1,197	539 (CU_SWE)	2,139 (UA_SWE)	-84	182
		10000-11000'	1,197	1,122	1,119 (UA_SWE)	1,351 (SNODAS)	50	-96

About SWEET

The following information provides a brief overview of SWEET, and the SWE products currently included in the ensemble framework. For more detailed descriptions, citations, and additional information, please refer to the GitHub repository.

SWE products currently used include:

1. **SNODAS SWE**
    a. The SNOw Data Assimilation System (SNODAS) is a physically based snow model developed by NOAA's National Operational Hydrologic Remote Sensing Center and operational since 2004. SNODAS produces daily 1-km gridded estimates of SWE and related snow variables by combining modeled weather inputs with satellite, airborne, and in situ snow observations.
2. **UA/SWANN SWE**
    a. The University of Arizona/Snow Water Artificial Neural Network Modeling System (UA/SWANN) is a research SWE product that uses snow models, artificial neural networks, and assimilated in situ measurements to generate daily gridded SWE and snow depth. The dataset provides 4-km and downscaled 800-m SWE across the conterminous United States, incorporating SNOTEL and COOP observations with PRISM precipitation and temperature fields.
3. **CU SWE**
    a. The University of Colorado (CU) SWE-Fusion product is an operational dataset produced by INSTAAR that provides near–real-time SWE estimates at 500-m resolution. The approach fuses SNOTEL observations with historical SWE reanalyses and terrain variables using a generalized linear model, and scales results using satellite-derived fractional snow-covered area from Snow Today.

The following notes describe key processing methods, assumptions, and visualization conventions used in generating SWE Evaluation Tool figures, tables, and elevation plots.

Figures:

- For ensemble figures, CU SWE and UA SWE are reprojected to the SNODAS resolution (1 sq km) and averages are only taken at pixels where at least 2 of the 3 products have SWE values greater than 0 inches.
- All overview and subbasin plots set pixels with values of 0 inches to NA so that the basemap is visible.
- For individual product figures, each product is displayed at its native resolution.
- Elevation bands of 250 feet are used in the elevation figures, and SWE volume is calculated for each product's native resolution in each band.
- Difference plots display pixels where snow was present in either the current or previous report.
- Figures are available in PNG format on the GitHub repository.

Tables:

- For all tables, mean, median, and volume are calculated at each product's native resolution - zeros are not removed from mean and median calculations.
- For ensemble tables, mean and median volume are calculated using the total volumes from the 3 products - not from an average aggregate raster.
- For min. and max., if 2 products have 0 volume, both products are listed in that cell. If all 3 products have 0 volume, no product is listed for min. or max. in that row.
- Change in SWE will only be calculated starting with the second report of the water year.
- Ensemble calculations: For HUC8 subbasins, the ensemble (mean and median) SWE and volume change (mean and median) are calculated individually for each subbasin. For the HUC6 "Total" row (tables 1 and 2), the ensemble is recalculated using the full HUC6 basin extent rather than summing HUC8. While the ensemble mean is additive and therefore consistent between HUC8 sums and the HUC6 total, the median reflects the middle value among the three products for the entire basin. As a result, median values will not sum to the HUC6 total, since they represent the difference between basin-wide medians rather than the sum of subbasin medians.
- Tables are available in CSV and PDF format on the GitHub repository.
- Elevation tables presented in the report (Tables 3, 4, and 5) are combined into one table (Table 3) on the GitHub repository.

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